KAPASA MAKASA UNIVERSITY

PLANT PROTECTION –NR 430

TEST 1

DATE: 22ND MARCH, 2022

DURATION: 1 HR, 30 MIN.

INSTRUCTIONS:

Answer ALL the questions in this paper

Use separate answer sheets

1. What do you understand by the following terms
   a. Virulence
   b. Epiphytes
   c. Facultative saprophytes
   d. Pathogenesis
   e. Phyto-anticipins **(10 marks)**

2. In details, discuss ''Disease Tetrahedron''. **(12 marks)**

3. State various methods that can be used to diagnose mollicutes infection in plants. **(4 marks)**

4. Describe how plants defend themselves against diseases using Induced Structural defence mechanisms. **(10 marks)**

5. When is the plant considered to be healthy? **(4 marks)**

Total= 40 marks

THE END..

1 (a) Virulence : This is the disease producing power of an organism. ✓ 1

(b) Epiphytes : These are plants which grow on other plants and obtain nutrients from them eg ~~mistletoe~~ dodder ✓ 2

(c) Facultative Saprophytes : These are fungi which grow on dead or decaying matter. e.g mushrooms 2

(d) Pathogenesis : ?

(e) phyto - anticipins ? 2

6

2) Disease Tetrahedron is a series of interconnected events which lead to disease development and pathogen propegation.

It involves four components and these are; environment, host plant, pathogen and time.

These components are dependant upon each other, one component can not succeed without the other components. We will look at each components, factors which needs to be in place for it to florish.

Pathogen: The pathogen in order for it to infect it requires certain conditions to be in place e..

Conditions

- Virulence: This is the disease producing power of an organism. for the pathogen to be able to cause an infection it need to have the disease causing power.
- Dispersal potential: A pathogen needs to be easily dispersed if it is to spread eg via wind, water, vectors e.t.c
- Host range: some pathogens require that the host plant be near in order for them to infect because they can not move on their own.
- Alternation of hosts: the ability of pathogens to cause disease between two or more plant families

Environment: this is various conditions which must be available in order for a pathogen to thrive and infect the host plant. these are;

Temperature: Each pathogen has an optimal temperature at which they operate. high temperature may halt pathogenic activities and very low temperatures respectively.

Light: If plants receive adequate amount of light, it means they will be dry and the physiological activities will be at the highest, meaning that there will be no room for pathogen ~~invasion~~ invasion

Soil type: ~~loam~~ loam soils favours pathogenic activities than clay soils. There is high pathogenic activities in light soils than soils with high organic matter.

Soil pH: The measure of acidity and alkalinity of soils. There are some pathogens which thrive in acidic soils, so it is better you lime the soils to reduce the acidic nature.

Air pollutants: pollutants can either aid or reduce diseases or the pathogens.

- Uniformity of host

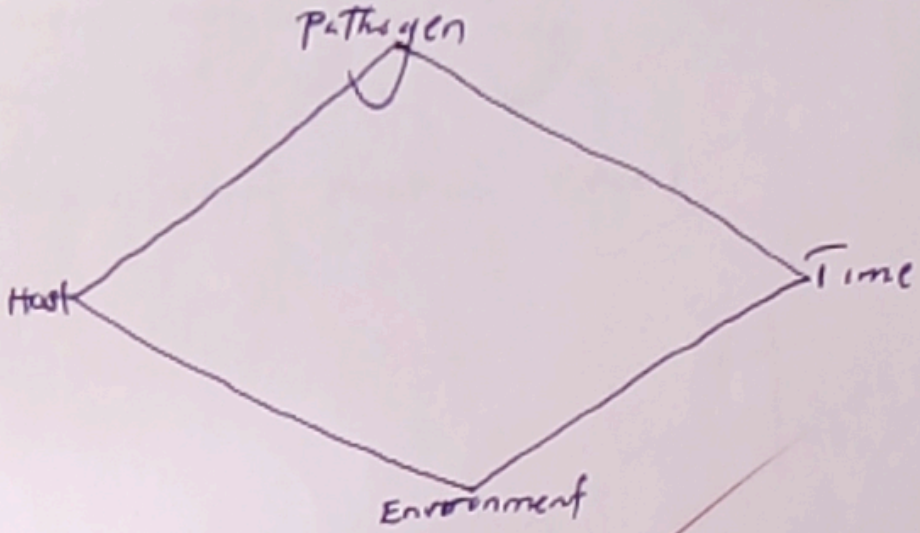
Host plant: This is the plant on which a pathogen would invade to cause disease. There are factors which need to be in place for the host to be protected from the pathogen and these are;

~~Age of the plant~~:

Seasonal development: This is the time/times ~~when~~ of the season when the plant is more or less susceptible to diseases

~~Host~~ Host resistance: This is the resistance of the host to the pathogen due to the plant not being the host to the pathogen

Time: This involves the duration of rainfall, seasonal changes e.t.c



12

3 Methods used to diagnose mollicutes infection in plants are;

- Enzyme linked immunosorbent Assay (ELISA)
- fluorescent antibody staining
- Grafting
- polymerase chain reaction (PCR)

4

4 plants defend themselves against diseases using induced structural defence mechanisms such as;

Development of hairy leaf and stem surface
this characteristics keeps pathogens away from landing on the plant and as a result protect or defends itself from being infected

Thorns: Some plants develop thorns either on the whole plant or around the leaves to keep away the pathogens from infecting or landing on them

(5) A plant is considered to be healthy when;

- Meristematic cells should divide and differentiate
- Different types of specialized cells should absorb water and nutrients from the soil to the aerial parts of the plant
- It is able to carry out photosynthesis, metabolize, translocate and store photosynthetic products
- It is able to produce seed and fruits or any other reproductive products

4

$\frac{27}{40} \times 100$

68%

The Copperbelt University

Kapasa Makasa University Campus

Department of Agriculture and Aquatic Sciences

SESSIONAL EXAMINATIONS 2022

COURSE CODE: NR 430

COURSE NAME: PLANT PROTECTION

TIME ALLOWED: THREE (3) HOURS

DATE: 19 TH OCTOBER 2022

INSTRUCTIONS:
1. Answer the COMPULSORY QUESTION ONE
2. Answer ONE other question from Section A to make it TWO
3. Answer at least ONE question from Section B or C
3. You are required to answer a total of five (5) questions in the exam as follows.
   i) 2 from section A including the COMPULSORY question one
   ii) At least ONE from BOTH section B and C
   iii) You are at liberty to pick ONE OR TWO from section A and Section B

NR 430 PLANT PROTECTION

SECTION A (Answer Question 1 and ONE other question from this section)

Question One (COMPULSORY)

a) Explain various parameters used in classification and identification of plant diseases. *(12 marks)*

b) Describe the various and distinct stages involved in pathogenesis. *(6 marks)*

c) What is dieback? *(2 marks)*

Question Two

a) Discuss disease tetrahedron in detail. *(12 marks)*

b) Define the following terms:

i. Virulence

ii. Phytopathogenesis

iii. Biotrophs

iv. Phytoanticipins

(8 marks)

Question Three

a) Explain the various cultural practices that aim at reducing the adverse effects of plant diseases. *(10 marks)*

b) Briefly explain how bacteria reproduce. *(5 marks)*

c) Explain the difference between viruses and viroids. *(5 marks)*

SECTION B (Answer one or two questions from this section)

Question Four

a) Pests are known to develop resistance to a certain pesticide after a certain period of time. Discuss how this phenomenon occurs. *(5 marks)*

b) Expalin how plants defend themselves against infections using pre-existing defense structures. *(10 marks)*

c) Explain how Enzyme Linked Immuno-sorbent Assay **(ELISA)** works as a molecular method of disease identification. *(5 marks)*

Question Five

a) Describe four (4) major plant viral diseases and how they affect a given plant species. *(10 marks)*

b) Briefly explain how Polymerase Chain Reaction **(PCR)** works as a plant disease identification method. *(5 marks)*

c) List the various symptoms and signs caused by plant bacterial diseases. *(5 marks)*

Section C (Answer one or two questions from this section)

Question Six

a) Discuss the various principles that anchor Integrated Pest Management (IPM). *(8 marks)*

b) Briefly explain various characteristics of arthropods *(8 marks)*

c) List various cultural practices that can be incorporated into an IPM system. *(4 marks)*

Question Seven

a) Insects are living organisms that belong to the Class *Insecta* and Phylum *Anthropoda*. State and explain five (5) insect "Orders" into which insects can be classified including examples of insect order you have selected. *(10 marks)*

b) Give a detailed explanation of the importance of understanding insect classification as an Agro-forester. *(10 marks)*